



Radio frequency identification (RFID) is used in many applications. Here, we present an access control system based on EM-18 RFID reader module and Arduino UNO board.

identification technology that uses radio signals to identify, track, sort and detect a variety of objects including people, vehicles, goods and assets without the need for direct contact or line-of-sight contact (as found necessary in bar code technology). RFID technology can track movement of objects through a network of radio-en-

energises the tag. The tag responds to the reader's query and announces its presence via radio waves, transmitting its unique identification data. This data is decoded by the reader and passed to the local application system via middleware. The middleware acts as an interface between the reader and the RFID application system. The system

then searches and matches the identity code with information stored in the host database or backend system. In this way, accessibility or authorisation for further processing can be granted or refused, depending on results received by the reader and processed by the database.

Circuit and working

Fig. 1 shows the circuit of RFID based access control using Arduino board.

The circuit is built on a UNO board (Board1), which includes an RFID module (RFID1), solenoid, and other components.

Arduino UNO board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software (called sketch). It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection

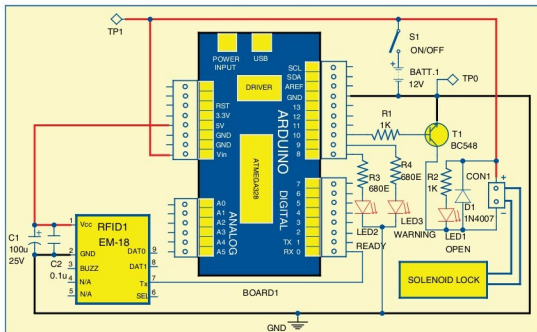


Fig. 1: Circuit diagram of the access control system

Semiconductors:

- | | |
|-----------|--------------------------|
| BOARD1 | - Arduino UNO board |
| RFID1 | - EM-18 RFID reader |
| T1 | - BC548 npn transistor |
| D1 | - 1N4007 rectifier diode |
| LED1-LED3 | - 5mm LED |

Resistors (all $\frac{1}{4}$ -watt, $\pm 5\%$ carbon):

- | | |
|--------|--------------|
| R1, R2 | - 1-kilo-ohm |
| R3, R4 | - 680-ohm |

Capacitors:

- C1 - 100 μ F, 25V electrolytic
C2 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1, CON2 - 2-pin terminal connector
S1 - On/off switch
BATT.1 - 12V battery
- 12V electric strike (solenoid lock)

abled scanning devices over a distance of several metres. A device called RFID tag, or simply a tag, is a key component of the technology.

An RFID system typically consists of three key elements:

1. An RFID tag, or transponder, that carries object-identifying data (unique ID code)
2. An RFID tag reader, or transceiver, that reads and writes tag data
3. A back-end database that stores records associated with tag contents

An RFID reader emits a low-level radio frequency magnetic field that